ScienceX

Battery Storage Market Notes, October 2026 issue

Science X Inc. Published November 1, 2026

"Grid-scale battery storage" (large batteries connected to the power grid that store electricity and release it) earns money by selling the "power" to fix electricity surpluses and shortages. It also earns from the way electricity prices change by time of day. These notes use published figures and our own records to show this month's movements in figures and tables.

1. Three income streams for a battery storage site (how to read these notes)

A battery storage site (grid-scale battery storage) earns income from three markets. ① The balancing market (EPRX) is where the "power" to fix electricity surpluses and shortages is sold to grid operators. ② Wholesale electricity (JEPX) is where electricity is bought and stored in cheap hours and sold in expensive hours. ③ The capacity market (OCCTO) is where the government buys future electricity supply capacity in advance. These notes set out the numbers in the order of these three markets. The ①②③ at the start of a section title shows which market it covers.

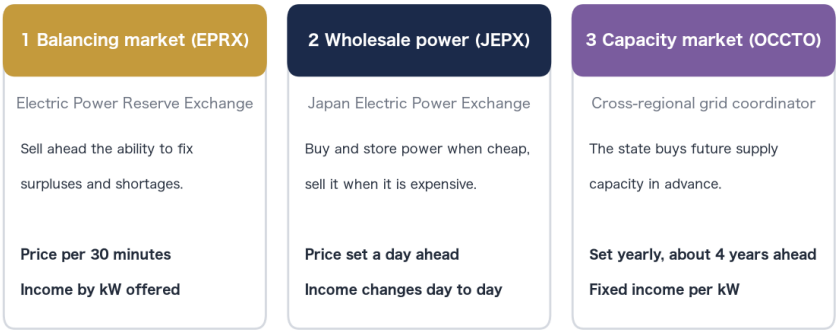


Figure 1 Three income streams for a battery storage site

Source: Rules of the Japan Electric Power Exchange (JEPX), the Electric Power Reserve Exchange (EPRX) and the Organization for Cross-regional Coordination of Transmission Operators (OCCTO).

2. Three trends this month

From the numbers in the three markets, we picked three movements this month.

- In the balancing market (EPRX), the bid ratio for Primary reserve (online) went above 100% in June 2026, at 106.8%. In September 2026 it was 123.1%, and the average winning price was 3.43 yen/kW per 30 min.
- In the balancing market (EPRX), in Tertiary reserve 2 in September 2026, the top 10% of 30-minute slots by revenue made up 56.9% of the month's revenue.
- In wholesale electricity (JEPX), in October 2026, the spread between the average of the 4 most expensive hours and the average of the 4 cheapest hours (monthly average) was 17.14 yen/kWh in Tokyo and 19.30 yen/kWh in Kyushu.

3. ① Balancing market (EPRX) | 7 products and prices by area

The balancing market is where grid operators buy the "power" to fix electricity surpluses and shortages every 30 minutes. Products are split by how fast output must change (Primary reserve acts on changes in frequency; Secondary and Tertiary reserve act on instructions from the grid operator). Primary reserve is the fastest, and Tertiary reserve 2 is the slowest. Primary reserve is a product that senses frequency changes by itself and acts. It has two categories: "online", where the grid operator monitors operation over a dedicated line, and "offline", where no dedicated line is installed and a record of operation is submitted afterwards. They are procured and traded separately, so we show them separately. A Composite product is a bid that bundles several products. Prices are set in yen per 1 kW of available power per 30 minutes. kW is a unit of how much power can be delivered, and is different from kWh, which measures an amount of electricity. One 30-minute trading round is called a "30-minute slot". The figure shows the range of winning prices by area (by the area where the resource is located). The thin line runs from the cheapest to the most expensive winning bid in the month, the thick band is the price range that holds 8 out of 10 slots with awards, and the dot is the average winning price. Nationwide numbers are in the tables in the "Market numbers" section. Published values for October 2026 are not yet in our totals, so we show the latest month with all products available: September 2026 (1st–29th) (preliminary values).

In the balancing market in September 2026, by area where the resource is located, the highest average winning price was 9.51 yen/kW per 30 min for Primary reserve (offline) in Tohoku, and the lowest was 0.38 yen/kW per 30 min for Tertiary reserve 2 in Shikoku.

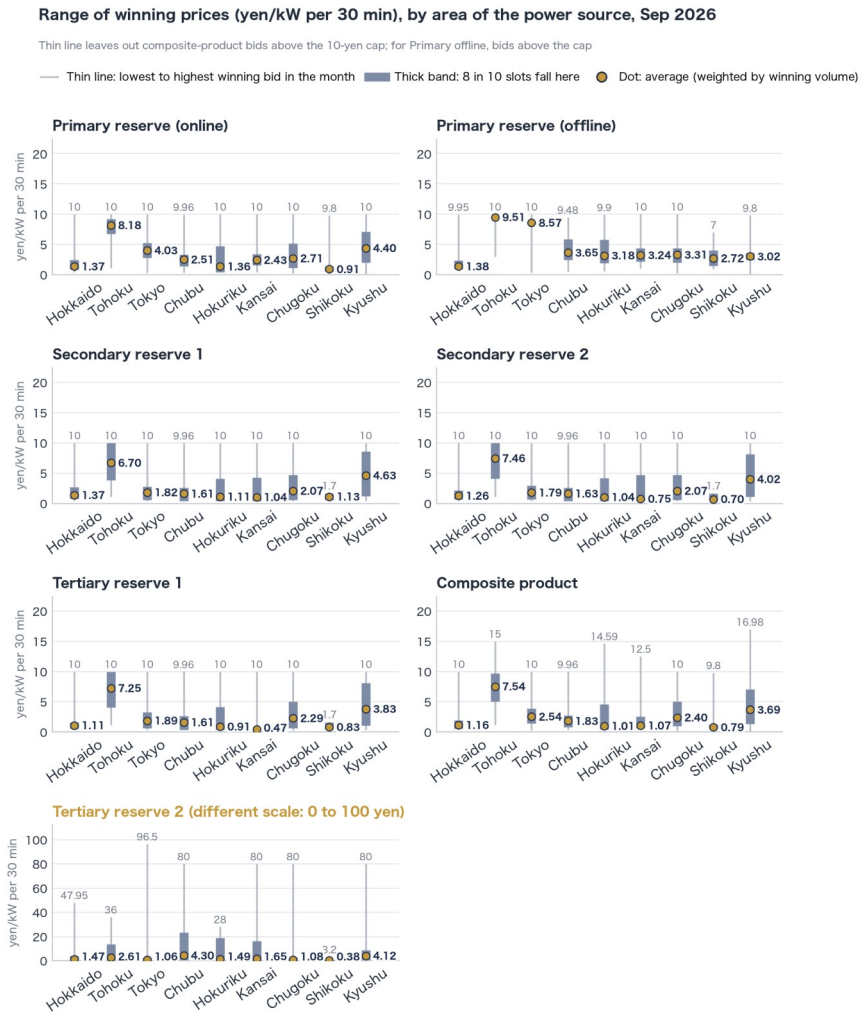


Figure 2 Range and average of winning prices by area in the balancing market (September 2026 (1st–29th), by area where the resource is located)

Source: Trading results (preliminary values) published by the Electric Power Reserve Exchange (EPRX). Our totals.

Product	Response time	Duration
Primary reserve (online)	Within 10 seconds	5 minutes or more
Primary reserve (offline)	Within 30 seconds	No set duration
Secondary reserve 1	Within 5 minutes	30 minutes
Secondary reserve 2	Within 5 minutes	30 minutes
Tertiary reserve 1	Within 15 minutes	30 minutes
Tertiary reserve 2	Within 60 minutes	30 minutes
Composite product	Requirements of the combined products	Requirements of the combined products

© Science X Inc. · https://www.scix.co.jp

4. ① Balancing market (EPRX) | Grid areas and times where bids are likely to win

The award rate is the share of bid volume that was awarded. The higher it is, the more likely bids are to win. The figure shows the award rate for each product by time of day (every 30 minutes), for each grid area (the area where the resource is located). Where a line breaks, there were no bids at that time.

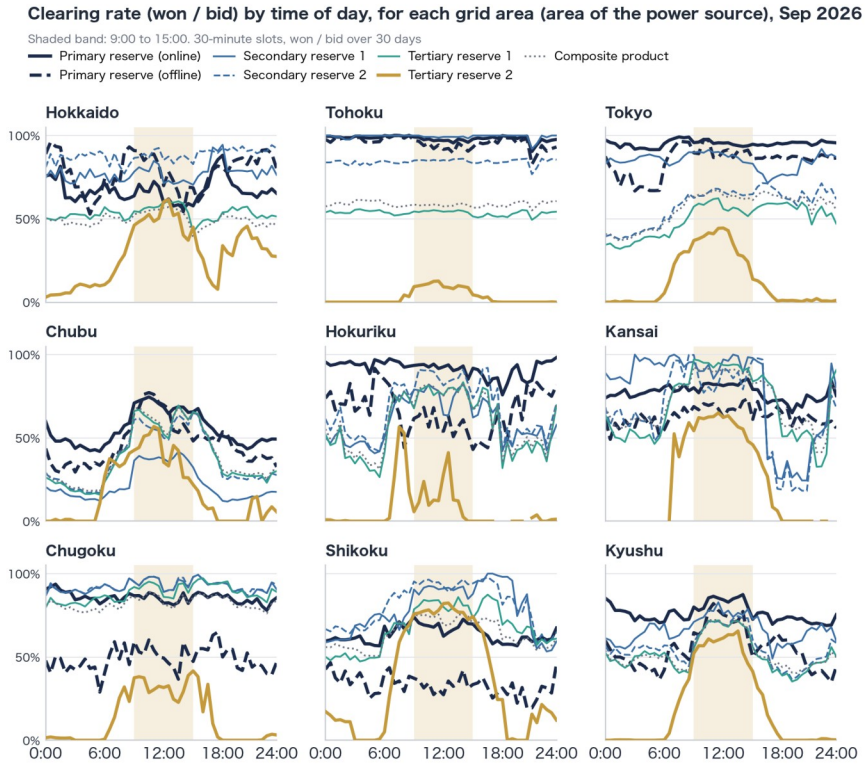


Figure 3 Award rate in the balancing market (grid area (area where the resource is located) × time of day, September 2026 (1st–29th))

Source: Trading results (preliminary values) published by the Electric Power Reserve Exchange (EPRX). Our totals.

5. ① Balancing market (EPRX) | Primary reserve (online) turned into a competition for places

Primary reserve is the fastest product. It responds to changes in frequency and changes output within 10 seconds (online) or within 30 seconds (offline). Batteries are often used for it. The left side is the online category and the right side is the offline category. The bars show the bid ratio (bid volume ÷ volume procured by grid operators). Above 100% means more bids came in than were procured. Even below 100%, some bids are not awarded (the award rate is in the "Market numbers" table). The bars rise when bids increase and also when procurement falls. The line is the average winning price. These are the monthly figures we have, from April 2025.

In the balancing market, the bid ratio for Primary reserve (online) went above 100% in June 2026, at 106.8%, and was 123.1% in September 2026. The average winning price went from 2.89 yen/kW per 30 min in April 2025 to 3.43 yen/kW per 30 min in September 2026, and the procured volume went from 3,066.1MW in April 2025 to 1,388.7MW in September 2026. For Primary reserve (offline), in September 2026 the bid ratio was 84.0% and the average winning price was 5.79 yen/kW per 30 min, and the procured volume went from 1,122.5MW in April 2025 to 497.8MW in September 2026.

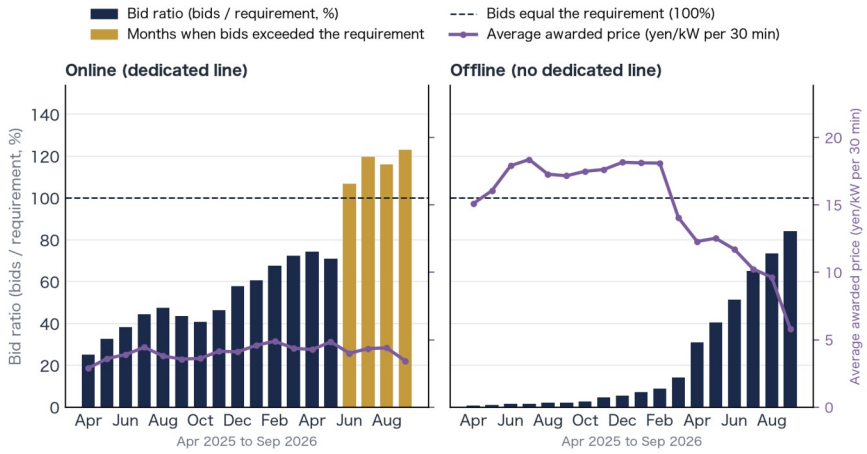


Figure 4 Bid ratio (bars) and average winning price (line) for Primary reserve (monthly; left is online, right is offline; gold bars are months when bids exceeded procurement)

Source: Trading results (preliminary values) published by the Electric Power Reserve Exchange (EPRX). Our totals.

6. ① Balancing market (EPRX) | How prices have moved by product

These are the average winning prices by product since April 2025 (monthly averages weighted by awarded volume). From March 14, 2026, Primary to Tertiary reserve 1 and the Composite product moved from 3-hour trading to 30-minute trading (Tertiary reserve 2 was already 30 minutes).

In the balancing market, from April 2025 to September 2026, average winning prices (yen/kW per 30 min) went from 2.89 yen to 3.43 yen for Primary reserve (online), from 15.10 yen to 5.79 yen for Primary reserve (offline), and from 1.12 yen to 2.22 yen for Tertiary reserve 2. The highest months were February 2026 at 4.90 yen for Primary reserve (online), July 2025 at 18.37 yen for Primary reserve (offline), and May 2026 at 4.18 yen for Tertiary reserve 2.

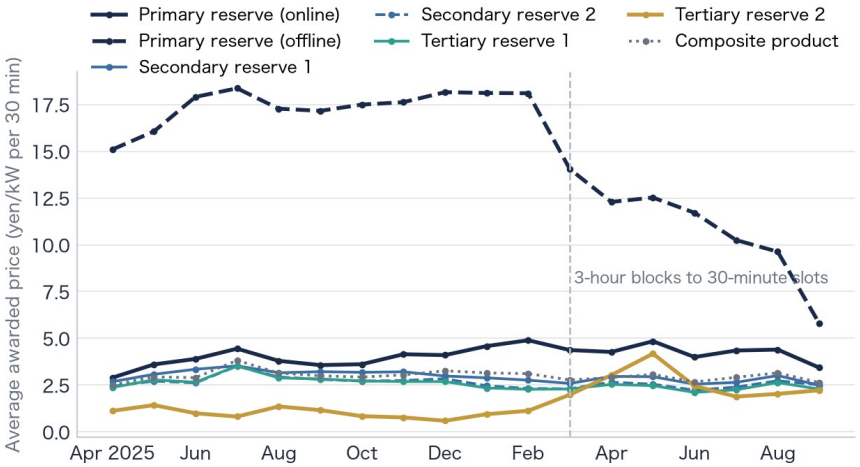


Figure 5 Average winning price by product in the balancing market (yen/kW per 30 min, monthly)
Source: Trading results (preliminary values) published by the Electric Power Reserve Exchange (EPRX). Our totals.

7. ① Balancing market (EPRX) | Which hours of the day are expensive

These are the hourly average winning prices by product for September 2026 (1st–29th).

In the balancing market in September 2026, average winning prices by hour were highest at 4.09 yen in the 18:00 hour and lowest at 2.75 yen in the 0:00 hour for Primary reserve (online), and highest at 5.02 yen in the 19:00 hour and lowest at 0.47 yen in the 0:00 hour for Tertiary reserve 2 (yen/kW per 30 min).

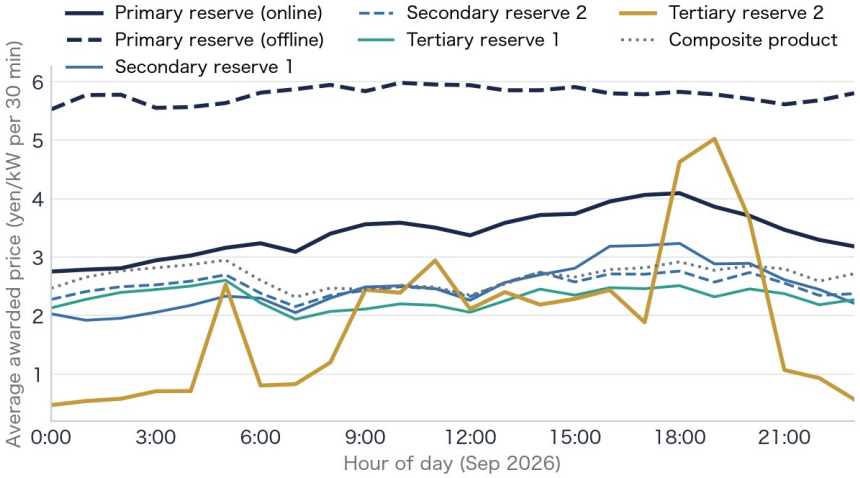


Figure 6 Average winning price by hour in the balancing market (September 2026 (1st–29th), yen/kW per 30 min)
Source: Trading results (preliminary values) published by the Electric Power Reserve Exchange (EPRX). Our totals.

8. ① Balancing market (EPRX) | Which products batteries are used in

The figure shows the share of awarded volume held by batteries. The table compares the average winning price of the volume awarded to batteries with the average winning price of the whole product.

In the balancing market in September 2026, the battery share was 27.9% for Primary reserve (online) and 100.0% for Primary reserve (offline). The average winning price of the volume awarded to batteries was 5.17 yen/kW per 30 min for Primary reserve (online) (3.43 yen/kW per 30 min overall), and 59.26 yen/kW per 30 min for Tertiary reserve 2 (2.22 yen/kW per 30 min overall).

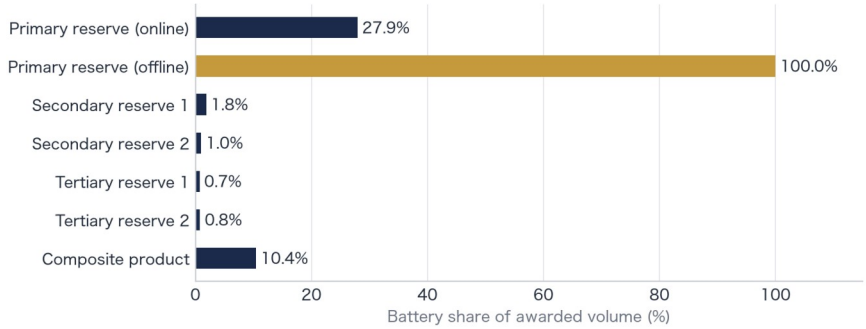


Figure 7 Share of batteries in the balancing market (%)

Source: Trading results by resource type (preliminary values) published by the Electric Power Reserve Exchange (EPRX). Our totals.

Product	Awarded to batteries (yen/kW per 30 min)	Whole product (yen/kW per 30 min)
Primary reserve (online)	5.17	3.43
Primary reserve (offline)	5.79	5.79
Secondary reserve 1	2.18	2.49
Secondary reserve 2	1.85	2.51
Tertiary reserve 1	1.86	2.28
Tertiary reserve 2	59.26	2.22
Composite product	5.13	2.64

Both are averages weighted by awarded volume (September 2026, preliminary values, nationwide). In Tertiary reserve 2, batteries make up only 0.8% of awarded volume. The nationwide average winning price in the slots where batteries won, weighted by battery awarded volume, is 5.82 yen. The battery average is high not because batteries are concentrated in expensive slots, but because the battery bids themselves are priced high (each winning bid is settled at its own bid price).

9. ① Balancing market (EPRX) | Revenue gathers in a small number of slots

We ranked the month's 30-minute slots with awards by revenue (average winning price × awarded volume, nationwide), from highest to lowest, and plotted for each product what share of the month's revenue the top slots hold. The more a curve bulges above the dotted line, the more revenue is concentrated in a few slots. Tertiary reserve 2 is a product that covers errors in forecasting renewable output caused by weather. Its price is low in most slots and high in only a few.

In the balancing market in September 2026, the top 1% and top 10% of slots by revenue held 10.1% and 56.9% of the month's revenue for Tertiary reserve 2, and 2.1% and 15.8% for Primary reserve (online).

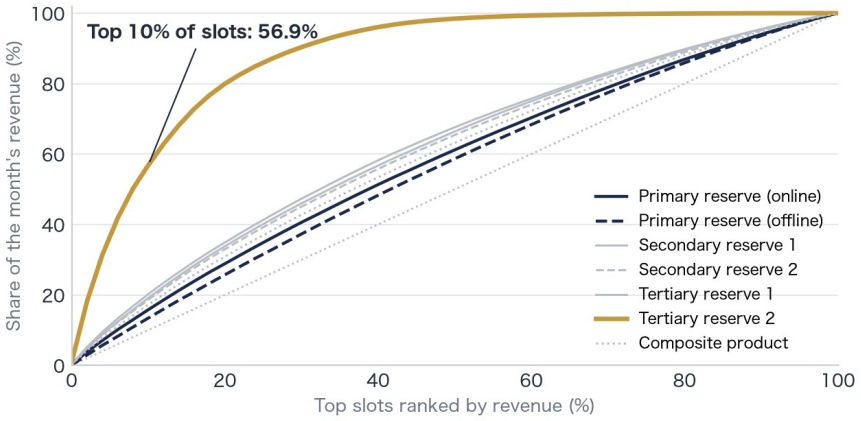


Figure 8 Share of revenue held by the top slots in the balancing market (September 2026 (1st–29th), nationwide)

Source: Trading results (preliminary values) published by the Electric Power Reserve Exchange (EPRX). Our totals.

In September 2026, out of 1,194 slots with awards, the top 1% of slots by revenue (average winning price × awarded volume) held 10.1% of the month's revenue. There were 25 slots with a nationwide average winning price of 10 yen/kW per 30 min or more, and 198 slots with no awards.

Slots with the highest top winning prices (by area where the resource is located). The highest winning price is the price of the single most expensive winning bid. The average winning price in the same slot and area is shown to the right:

Day	Time	Area	Highest winning price (yen/kW per 30 min)	Average winning price in the same slot and area (yen/kW per 30 min)
September 8	9:30	Tokyo	96.50	6.01
September 8	10:30	Tokyo	96.50	6.28
September 8	11:00	Tokyo	96.50	6.56
September 27	11:00	Tokyo	96.00	5.82
September 27	11:30	Tokyo	96.00	5.18

Source: Tertiary reserve 2 trading results (preliminary values) published by the Electric Power Reserve Exchange (EPRX). Our totals.

10. ② Wholesale electricity (JEPX) | Average prices

Wholesale electricity is the electricity that power companies and retailers buy and sell. Its price is set in the "spot market" of the Japan Electric Power Exchange (JEPX) (the market where electricity for the next day is traded the day before). "System (nationwide)" is the price when all of Japan is treated as one market (the system price). When transmission lines between regions get congested, prices split by area from Hokkaido to Kyushu (area prices).

The monthly average of the wholesale electricity (JEPX) System (nationwide) price was 22.51 yen/kWh in October 2026 and 11.67 yen/kWh in the same month last year, a difference of 10.84 yen.

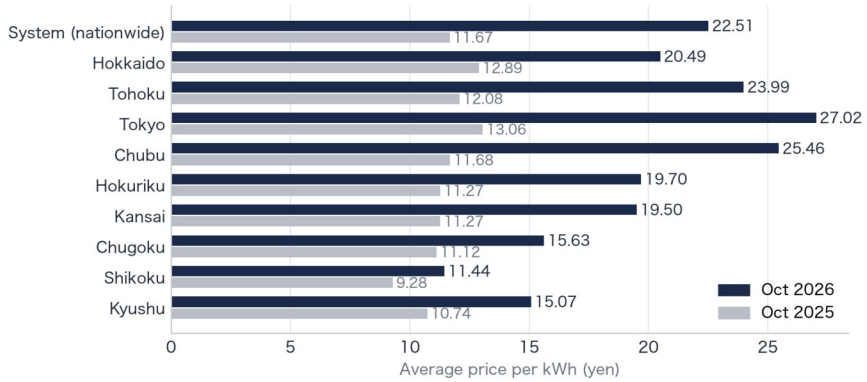


Figure 9 Average wholesale electricity price (yen/kWh). Navy is October 2026, grey is the same month last year

Source: Spot market trading results published by the Japan Electric Power Exchange (JEPX). Simple average of 30-minute prices. Our totals.

11. ② Wholesale electricity (JEPX) | How prices move during the day

Wholesale electricity prices change every 30 minutes. The figure shows the daily shape, averaging prices at the same time of day across the month. If a battery stores electricity in cheap hours and releases it in expensive hours, the difference becomes its income.

In October 2026, average wholesale electricity prices during the day ranged from a low of 19.03 yen around 12:00 to a high of 37.11 yen around 16:30 in Tokyo, and from 4.94 yen around 12:00 to 24.74 yen around 17:30 in Kyushu.

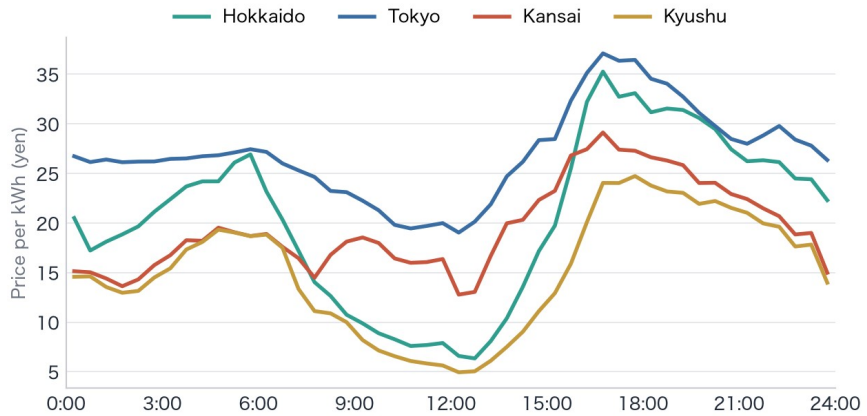


Figure 10 Daily peaks and troughs of wholesale electricity prices (October 2026 average, yen/kWh)

Source: JEPX spot market. Our totals.

12. ② Wholesale electricity (JEPX) | How large the spread is per kWh

We took the average price of the day's 4 most expensive hours minus the average price of the 4 cheapest hours, and show the monthly average by area. It equals the difference per kWh when you buy 1 kWh in the 4 cheapest hours and sell it in the 4 most expensive hours. It is a simple difference: electricity lost in charging and discharging and fees are not deducted, and the order of cheap and expensive hours is ignored.

In October 2026, the spread between the 4 most expensive and 4 cheapest hours of wholesale electricity (monthly average) was largest in Hokkaido at 29.02 yen/kWh and smallest in Tokyo at 17.14 yen/kWh.

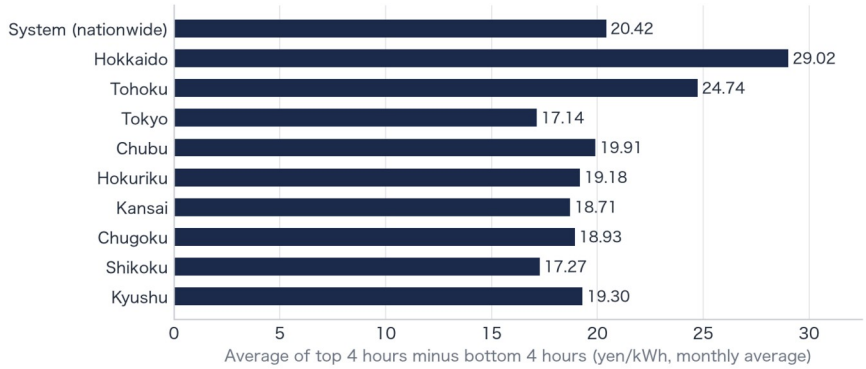


Figure 11 Wholesale electricity spread: average of the 4 most expensive hours − average of the 4 cheapest hours (October 2026, yen/kWh, monthly average)

Source: JEPX spot market. Our totals.

Area	2-hour spread (yen/kWh)	4-hour spread (yen/kWh)	Days with a 4-hour spread of 10 yen or more
System (nationwide)	22.95	20.42	29
Hokkaido	32.56	29.02	31
Tohoku	27.09	24.74	30
Tokyo	19.62	17.14	29
Chubu	22.37	19.91	29
Hokuriku	21.71	19.18	31
Kansai	21.22	18.71	31
Chugoku	20.90	18.93	31
Shikoku	19.40	17.27	28
Kyushu	21.19	19.30	31

The spread is the monthly average of the daily spreads. October 2026 covers 31 days.

13. ② Wholesale electricity (JEPX) | There is a spread every day

We show the same 4-hour spread day by day. The bigger the spread on a day, the more room a battery has to earn from it.

Of the 31 days in October 2026, the wholesale electricity 4-hour spread was 10 yen or more on 29 days for System (nationwide), 31 days for Hokkaido, Hokuriku, Kansai, Chugoku and Kyushu, and 28 days for Shikoku.

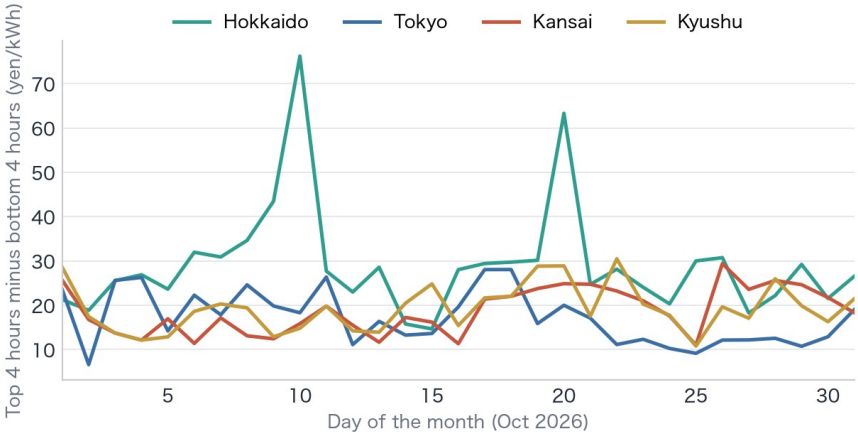


Figure 12 Daily wholesale electricity spread (the day's average of the 4 most expensive hours − average of the 4 cheapest hours, yen/kWh)

Source: JEPX spot market. Our totals.

14. ③ Capacity market (OCCTO) | A market that buys future supply capacity

The capacity market buys electricity supply capacity four years ahead through an auction held once a year. Prices are set in yen per kW per year. The left figure shows clearing prices in past main auctions. The right figure shows the battery storage and pumped hydro capacity contracted in the Long-Term Decarbonization Power Source Auction (a separate auction that promises capacity income, in principle for 20 years, for things like building new decarbonized power sources). Results are published once a year, so they do not change every month. The latest is for delivery FY2029 (published January 2026).

In the capacity market main auction, the clearing price for Tokyo was 14,137 yen/kW for delivery FY2024, 3,495 yen/kW for FY2025, 5,834 yen/kW for FY2026, 9,555 yen/kW for FY2027, 14,812 yen/kW for FY2028 and 15,111 yen/kW for FY2029. In the Long-Term Decarbonization Power Source Auction, the contracted capacity of battery storage and pumped hydro was 1,669 MW in the FY2023 auction; in the FY2024 auction, 961 MW for a duration of 3 hours to under 6 hours and 769 MW for 6 hours or more; and in the FY2025 auction, 819 MW for lithium-ion battery storage and pumped hydro (excluding new builds) and 886 MW for non-lithium-ion battery storage, pumped hydro (new builds) and LDES. Both are annual values, so these are the latest published figures and do not change every month.

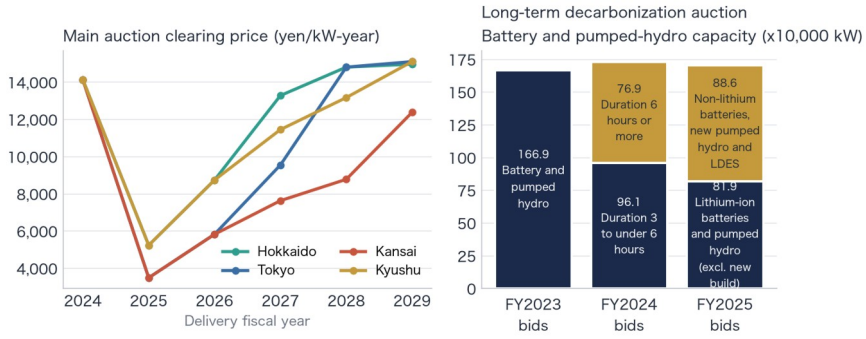


Figure 13 Capacity market clearing prices (left, yen/kW) and contracted capacity of battery storage and pumped hydro in the Long-Term Decarbonization Power Source Auction (right, in units of 10,000 kW)

Source: Clearing results and capacity contract documents published by the Organization for Cross-regional Coordination of Transmission Operators (OCCTO) (sources listed in the last section (Assumptions and terms)). As published.

© Science X Inc. · https://www.scix.co.jp

15. Our records | Where battery storage projects are increasing

We counted the projects newly listed this month on DealRoom (our data room for buyers, where we list battery storage projects), by grid area of the grid operator. These are numbers for projects we handled, not counts for the whole market. Transfer of rights is a sale that hands over the land and grid rights. Completed-build handover is a sale that hands over the site once the equipment is built.

New listings on DealRoom in October 2026 were 77 projects and 309.1 MW. By grid area, these included Tokyo with 33 projects and 99.8 MW, Tohoku with 8 projects and 73.4 MW, Chubu with 11 projects and 69.9 MW, and Kyushu with 13 projects and 19.9 MW.

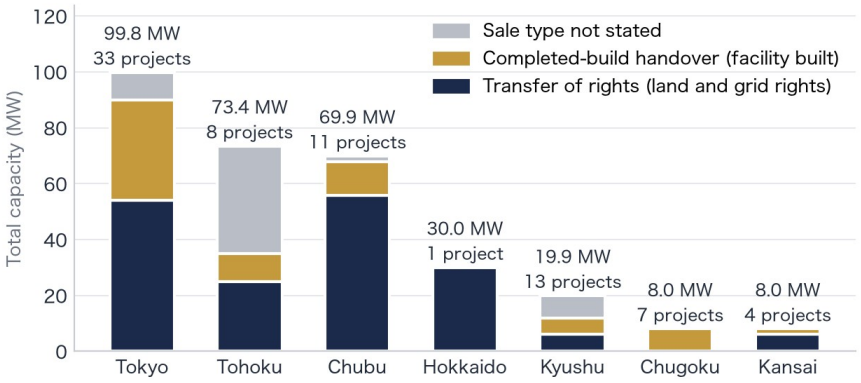


Figure 14 Output (MW) of projects we listed on DealRoom in October 2026

Source: Our DealRoom records (project locations are shown only down to the grid area; prices and parties are not shown).

16. Our records | How many months and how much it costs to connect to the grid

A grid connection study is the process of asking the grid operator to check "can we connect at this site, and how much will the work cost". The grid connection cost contribution is the part of the connection work cost paid by the applicant. Of the response letters we have received, we totaled only the high-voltage ones (connecting to 6,600V distribution lines), by grid area. Extra-high voltage is not included. Grid areas with fewer than 5 cases are not shown.

The medians of grid connection study responses (high voltage only, as of October 19, 2026) ranged from 8.0 months in Tokyo to 17.0 months in Tohoku for construction time, and from ¥3.02 million in Tokyo to ¥32.35 million in Tohoku for the grid connection cost contribution (Tohoku: n=5 and n=8).

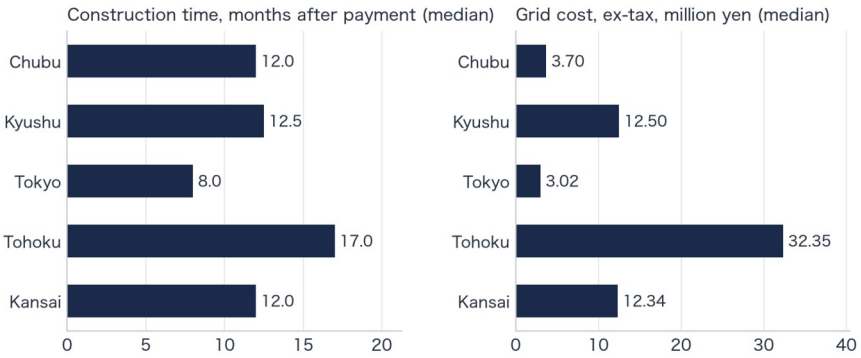


Figure 15 Medians of grid connection study responses (high voltage) by grid area (left: months of construction work, right: grid connection cost contribution, excluding tax)

Source: Our totals of the high-voltage portion of grid connection study response letters we received from grid operators (as of October 19, 2026).

17. Market numbers (tables you can use as they are)

We put the numbers behind the figures into tables you can use in your own documents: ① nationwide numbers by product in the balancing market (bid ratio, award rate, price, battery share), ② wholesale electricity (JEPX) numbers by area and trading volume, and ③ capacity market clearing prices.

Balancing market (September 2026, preliminary values, nationwide)

Product	Procured volume (MW)	Bid volume (MW)	Awarded volume (MW)	Bid ratio (bids ÷ procurement)	Award rate (awarded ÷ bids)	Average winning price (yen/kW per 30 min)	Battery share
Primary reserve (online)	1,388.7	1,710.0	1,337.3	123.1%	78.2%	3.43	27.9%
Primary reserve (offline)	497.8	418.4	265.7	84.0%	63.5%	5.79	100.0%
Secondary reserve 1	962.6	1,996.2	1,102.3	207.4%	55.2%	2.49	1.8%
Secondary reserve 2	983.1	3,709.3	2,163.2	377.3%	58.3%	2.51	1.0%
Tertiary reserve 1	2,498.5	5,895.9	3,021.0	236.0%	51.2%	2.28	0.7%
Tertiary reserve 2	376.3	1,371.7	368.9	364.5%	26.9%	2.22	0.8%
Composite product	3,383.8	6,641.4	3,623.5	196.3%	54.6%	2.64	10.4%

Volumes are averages per 30-minute slot. For products with a bid ratio below 100%, bids fall short of procurement. The award rate is the share of bid volume that was awarded.

Source: Trading results (preliminary values) and trading results by resource type published by the Electric Power Reserve Exchange (EPRX). Our totals.

Wholesale electricity, JEPX spot market (October 2026, by area, yen/kWh)

Area	Monthly average	Same month last year	Daytime (8:00–16:00)	Evening (16:00–20:00)	4-hour spread	Highest	Slots at 30 yen or more	Slots at 0.01 yen
System (nationwide)	22.51	11.67	16.02	31.06	20.42	47.88	220	39
Hokkaido	20.49	12.89	11.31	32.25	29.02	105.00	303	67
Tohoku	23.99	12.08	15.78	34.19	24.74	50.01	437	95
Tokyo	27.02	13.06	23.12	34.69	17.14	50.01	466	16
Chubu	25.46	11.68	20.38	33.55	19.91	46.45	387	24
Hokuriku	19.70	11.27	18.29	26.92	19.18	46.45	218	24
Kansai	19.50	11.27	18.22	26.76	18.71	46.45	215	29
Chugoku	15.63	11.12	9.86	23.10	18.93	44.79	63	79
Shikoku	11.44	9.28	7.50	16.61	17.27	37.51	23	146
Kyushu	15.07	10.74	8.31	23.10	19.30	44.79	53	91

Trading volume (October 2026, total of 31 days): total contracted volume 30,319 million kWh (same month last year 22,279 million kWh), sell bids 45,218 million kWh, buy bids 38,191 million kWh.

Source: Spot market trading results published by the Japan Electric Power Exchange (JEPX). Prices are simple averages of 30-minute slots. Our totals.

Capacity market main auction area prices (yen/kW per year)

Delivery fiscal year	Hokkaido	Tohoku	Tokyo	Chubu	Hokuriku	Kansai	Chugoku	Shikoku	Kyushu
FY2024	14,137	14,137	14,137	14,137	14,137	14,137	14,137	14,137	14,137
FY2025	5,242	3,495	3,495	3,495	3,495	3,495	3,495	3,495	5,242
FY2026	8,749	5,833	5,834	5,832	5,832	5,832	5,832	5,832	8,748
FY2027	13,287	9,044	9,555	7,823	7,638	7,638	7,638	7,638	11,457
FY2028	14,812	14,812	14,812	10,280	8,785	8,785	8,785	8,785	13,177
FY2029	14,972	15,111	15,111	12,388	12,388	12,388	12,388	12,388	15,112

Contracted capacity of battery storage and pumped hydro in the Long-Term Decarbonization Power Source Auction (in units of 10,000 kW)

Auction fiscal year	Category	Contracted capacity (10,000 kW)
FY2023 auction	Battery storage and pumped hydro	166.9
FY2024 auction	Duration of 3 hours to under 6 hours	96.1
FY2024 auction	Duration of 6 hours or more	76.9
FY2025 auction	Lithium-ion battery storage and pumped	81.9

© Science X Inc. · https://www.scix.co.jp

Auction fiscal year	Category	Contracted capacity (10,000 kW)
	hydro (excluding new builds)	
FY2025 auction	Non-lithium-ion battery storage, pumped hydro (new builds) and LDES	88.6

Source: Clearing results and capacity contract documents published by the Organization for Cross-regional Coordination of Transmission Operators (OCCTO) (sources listed in the last section (Assumptions and terms)). As published.

18. Assumptions and terms

- The numbers are our totals of the values published by the sources. We do not add up several months or convert to annual figures. We do not show revenue amounts for any single battery storage site.
- Months covered: the balancing market (EPRX) uses preliminary values for September 2026 (delivery days up to the 29th; up to the 30th for Primary reserve (offline)), and wholesale electricity (JEPX) uses October 2026 (31 days). The balancing market covers a different month until the published values for October 2026 are in our totals.
- Balancing market: EPRX preliminary values, nationwide (total column). Procured volume is by grid operator (TSO); bid volume, awarded volume and average winning price are by area where the resource is located. Volumes are averages per 30-minute slot. The average winning price is weighted by awarded volume. Bid ratio = bid volume ÷ procured volume; award rate = awarded volume ÷ bid volume. For 3-hour blocks up to March 13, 2026, we counted each block as six 30-minute slots.
- Range of winning prices: the thin line runs from the cheapest bid (lowest winning price) to the most expensive bid (highest winning price) among the month's slots with awards (published values by area where the resource is located). The published values for Primary reserve (online), Secondary reserve 1 and 2, and Tertiary reserve 1 also include Composite product bids (one price covering several products), so we removed from the thin line bids above 10 yen that can be explained by Composite product bids in the same slot and area. For Primary reserve (offline), we removed from the thin line bids above the price cap (10 yen this month) (the balancing market system awards bids above the cap too, and cuts them down to the cap at settlement). The thick band is the 10% to 90% range when the average winning prices of slots with awards are sorted from cheapest. The dot is the average weighted by awarded volume.
- Award rate by grid area × time of day: total awarded volume ÷ total bid volume for slots at the same time of day in the month (by area where the resource is located).
- The response time and duration of balancing market products are the FY2026 specifications (Electric Power Reserve Exchange, "Product Requirements", 6th edition). Primary reserve senses frequency changes by itself and acts. Online is the category where the grid operator monitors operation over a dedicated line, and offline is the category where no dedicated line is installed and a record of operation is submitted afterwards. The offline response time is within 30 seconds, and there is no set duration.
- Expensive slots in Tertiary reserve 2: slots with a nationwide average winning price of 10 yen/kW per 30 min or more. The top 1% is the top 1% of slots ranked by revenue (average winning price × awarded volume) (1% of the number of slots, rounded). The highest winning price is the published value by area where the resource is located, and is the price of the single most expensive winning bid. In the balancing market, each winning bid is settled at its own bid price (pay-as-bid (multi-price)), so the average winning price in the same slot and area is lower than this.
- Average prices are simple averages of 30-minute slots (not weighted by trading volume). Daytime is the slots from 8:00 to 16:00, and evening is the slots from 16:00 to 20:00.
- Spread: we sort the day's 30-minute slots by price and subtract the average of the 8 cheapest slots from the average of the 8 most expensive slots (4 hours) (4 slots for 2 hours). We do not consider the order of cheap and expensive hours, electricity lost in charging and discharging, or fees, taxes and transmission charges. It does not show actual revenue.

Sources (OCCTO, capacity market clearing results): delivery FY2024 https://www.occto.or.jp/iinkai/youryou/kentoukai/2020/files/youryou_kentoukai_27_05.pdf / delivery FY2025 https://www.occto.or.jp/iinkai/youryou/kentoukai/2021/files/youryou_kentoukai_35_03_1.pdf / delivery FY2026 https://www.occto.or.jp/iinkai/youryou/kentoukai/2022/files/youryou_kentoukai_43_03.pdf / delivery FY2027 https://www.occto.or.jp/market-board/market/oshirase/2023/files/240124_mainauction_youryouyakujokekka_kouhyou_jitsujukyu2027.pdf / delivery FY2028 https://www.occto.or.jp/market-board/market/oshirase/2024/files/250129_mainauction_youryouyakujokekka_kouhyou_jitsujukyu2028.pdf / delivery FY2029 https://www.occto.or.jp/assets/news/capacity-market/260120_mainauction_youryouyakujokekka_kouhyou_jitsujukyu2029.pdf

Sources (OCCTO, Long-Term Decarbonization Power Source Auction): FY2023 auction https://www.occto.or.jp/assets/260605_auction_keiyakujyoukyou_kouhyou.pdf / FY2024 auction https://www.occto.or.jp/market-

board/market/jitsujukyukanren/files/longauction_keiyakukekka_2024.pdf / FY2025 auction
https://www.occto.or.jp/assets/various/capacity-market/jitsujukyukanren/yoryokakuhokeiyaku/longauction_keiyakukekka_2025.pdf

- Capacity market clearing results for each delivery fiscal year are published in a different year (results for FY2024 in 2020, for FY2029 in January 2026). We do not show the number of battery projects in the Long-Term Decarbonization Power Source Auction, because it is not in the published summary tables.

Sources: Japan Electric Power Exchange JEPX spot market trading results / Electric Power Reserve Exchange EPRX trading results (preliminary values) and trading results by resource type / Organization for Cross-regional Coordination of Transmission Operators (OCCTO) capacity market main auction clearing results and Long-Term Decarbonization Power Source Auction capacity contract documents / our totals of grid connection study response letters we received / our DealRoom records.

Terms: System price (System, nationwide) = the wholesale electricity price when all of Japan is treated as one market. Area price = the price that splits by area depending on congestion on transmission lines. The grid = the power companies' transmission network. Grid connection = connecting to the transmission network. MW = the size of power output (1MW = 1,000kW). MWh = the amount of electricity that can be stored. Yen/kWh = the price per 1kWh of electricity.

Science X Inc. https://www.scix.co.jp Contact: s@scix.co.jp

Newsletter "Japan grid-scale BESS: primary data":
https://www.linkedin.com/newsletters/7521830530569445376/

You may use the figures and numbers if you cite the source (citation guide: https://www.scix.co.jp/press).